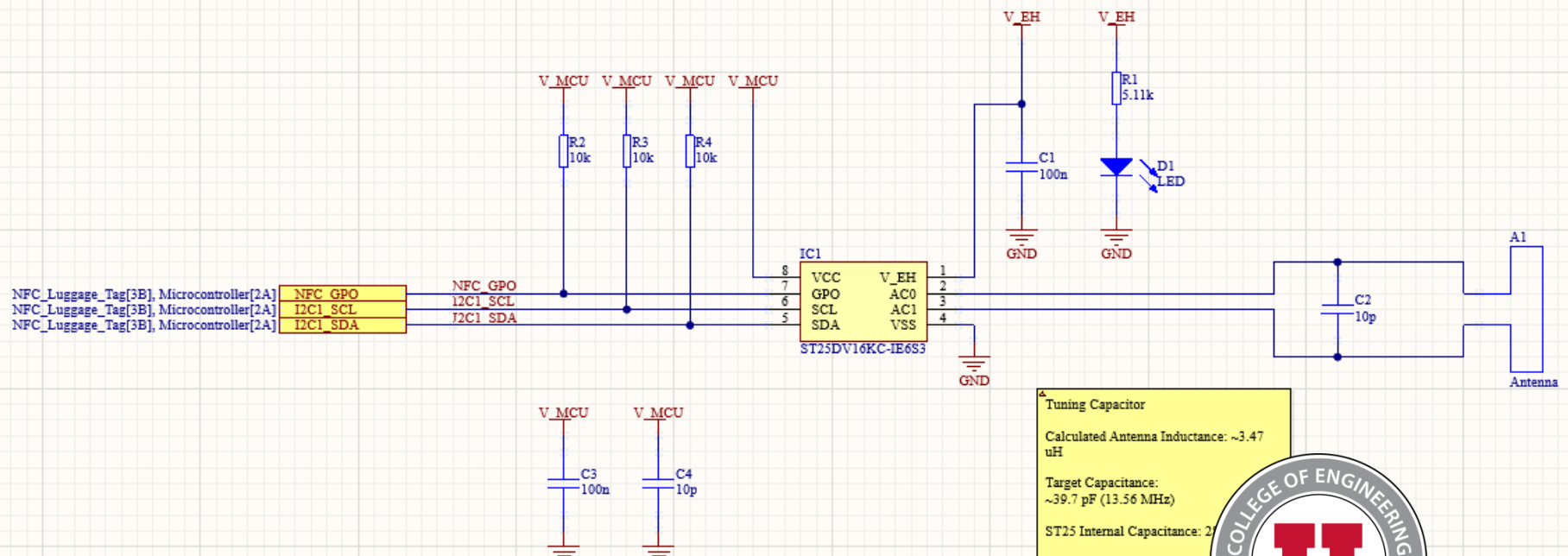
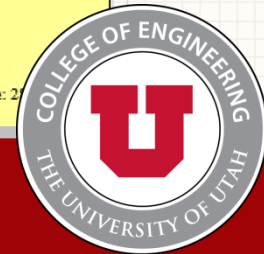


IEEE PCB Design Workshop: (Week 01) Introduction to Schematics



Adrian Sucahyo and IEEE at the University of Utah
Department of Electrical and Computer Engineering
University of Utah, Salt Lake City, UT



Workshop Outline

Tentatively scheduled for 6-7 weeks

- Sept. 15 – Introduction to Schematics (WEB L102)
- Sept. 22 – Schematics Continued (WEB 114)
- Sept. 29 – Introduction to PCB Layout (WEB L102)
- Oct. 6 – Layout Continued (WEB L102)
- Oct. 8 PCBs DUE!
- **** FALL BREAK ****
- TBD – Soldering Week 1
- TBD – Soldering Week 2
- TBD – Soldering Week 3 (Optional)

Workshop Logistics

- We **highly** recommend attending all workshop sessions
 - Each session is designed to build off previously discussed materials
- By the end of the workshop, you will be able to:
 - Build an NFC Luggage Tag
 - Understand the basics of schematic layout, PCB layout, and board manufacturing
 - Gain experience with through-hole and surface mount PCB assembly
 - Have a basic understanding of manufacturing techniques and equipment used in the industry

Workshop Logistics

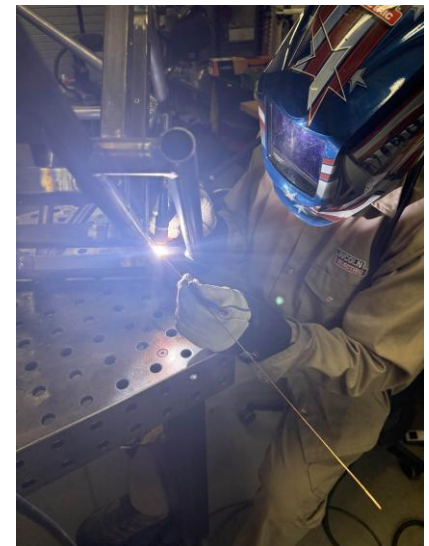
- We hope to not charge a fee for this workshop and supporting materials
 - Tentative on ASUU funding for the Fall Semester
 - Will know by October if fees will be necessary
 - If fees are necessary to support the workshop, the cost will be as minimal as possible
- Want to pursue a different project? Chat or Email Us!
 - We may be able to support individual projects get board manufactured

Want more experience?

- Consider joining the FSAE tractive team!
 - The Tractive Team is currently looking for students to assist with designing and assembling the electrical system for an electric formula-style race car!
 - No experience required!

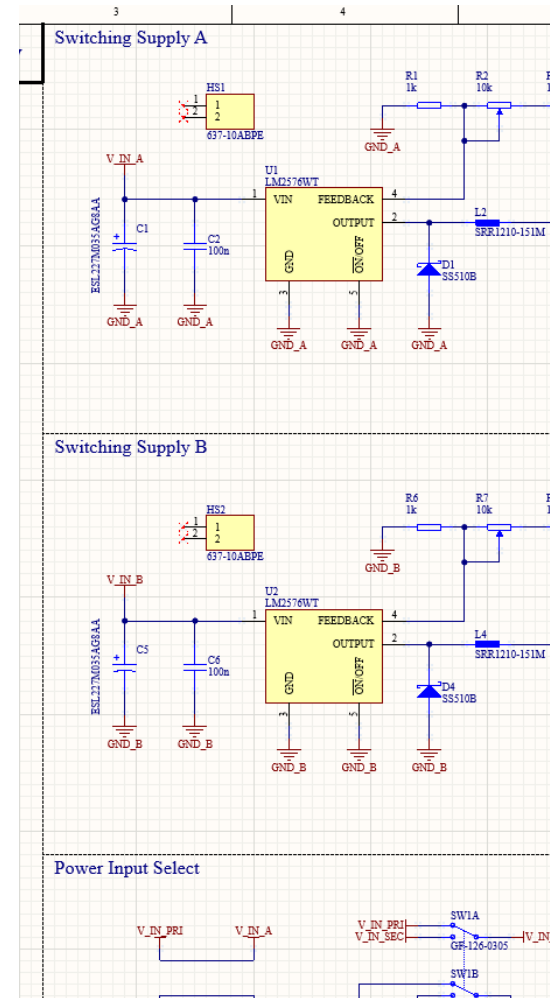


U of U FSAE Discord Link



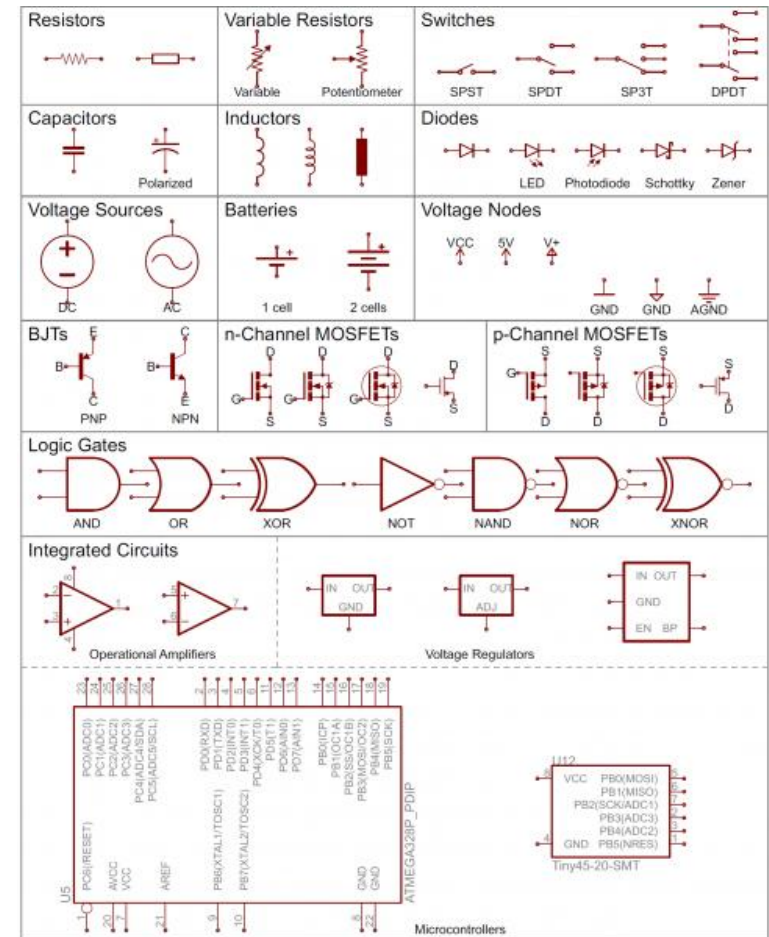
What is a schematic?

- Schematics are a graphical representation of an electrical circuit or system
 - A step up from a block diagram
 - Shows all parts and electrical connections in a circuit
 - Used to route PCB traces during layout
- Schematics are NOT necessarily a physical representation of component positions on the PCB



Components in a Schematic

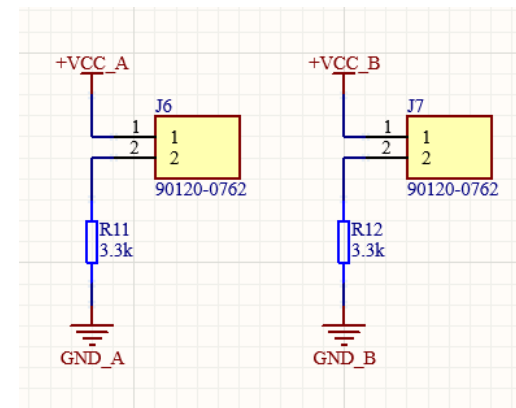
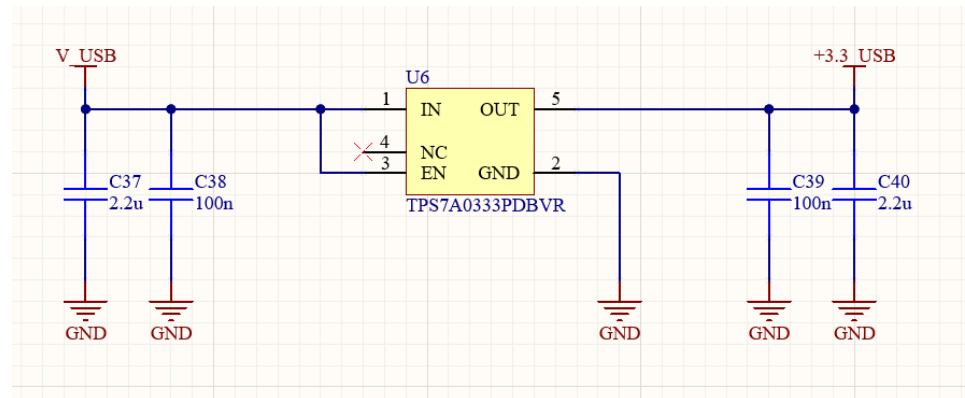
- Every electrical component will have its own symbol within the schematic
 - Contains all pins connected to the component
 - May contain additional information such as signal type
 - Relatively consistent across platforms



<https://learn.sparkfun.com/tutorials/how-to-read-a-schematic/all>

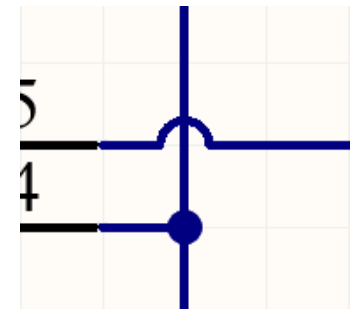
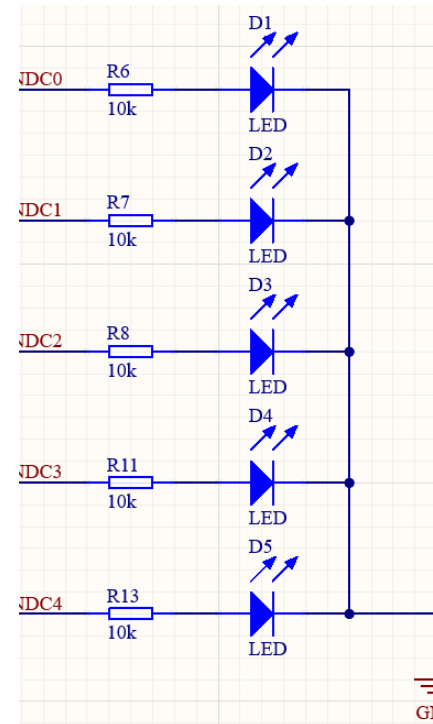
Symbol Attributes

- Symbols may have additional information regarding the component they represent
 - Designators
 - Values
 - Part Numbers
 - Pin Name
 - Pin Number



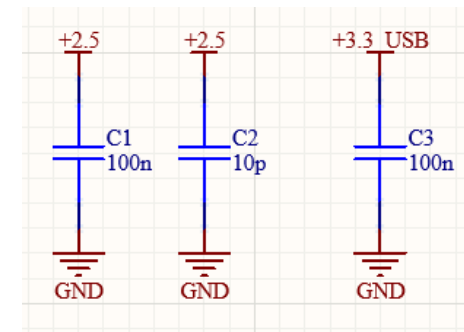
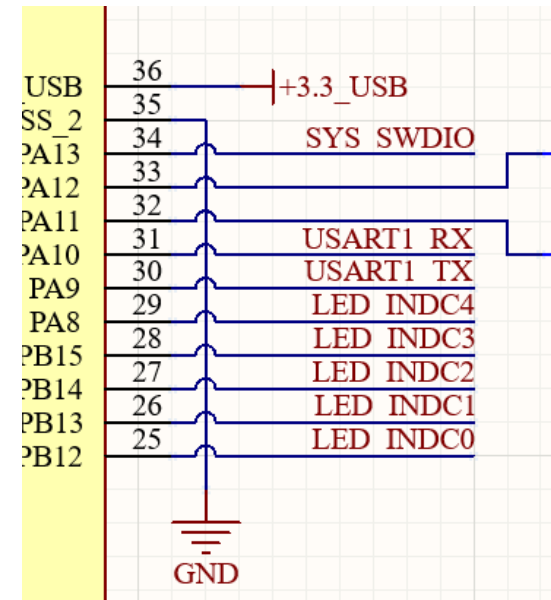
Nets and Nodes

- Schematic Nets are how components are electrically connected in a circuit
 - Represented with lines between components
- Nodes/Junctions are locations where wires split in two (or more)
 - A connection on a node is denoted by a dot



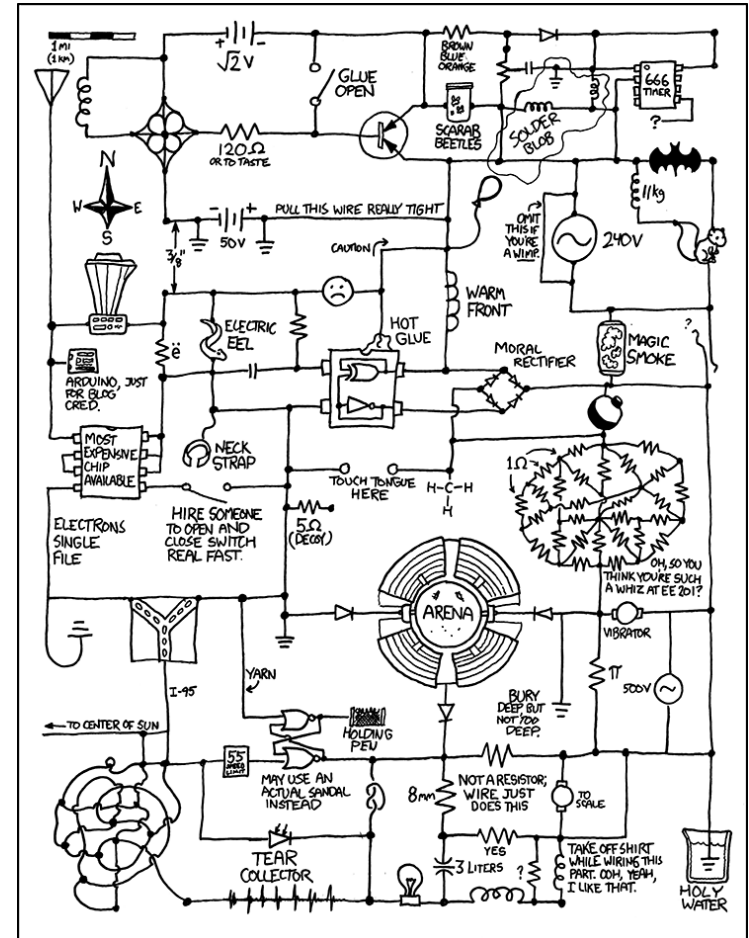
Net Labels

- Net Labels allow electrical connections to be made without having to draw a line between components
 - Net Labels tend to be local to the current schematic sheet
 - Power Labels tend to be global to the full design
- Note: Ports are similar but have a different purpose
 - Useful for hierarchal / multi-sheet design



Good Schematic Practice

- Readability is KEY!
 - The schematic needs to be read by other engineers to assist with layout and debugging
 - Readable schematics make it easier to find electrical errors which would be translated into the physical layout

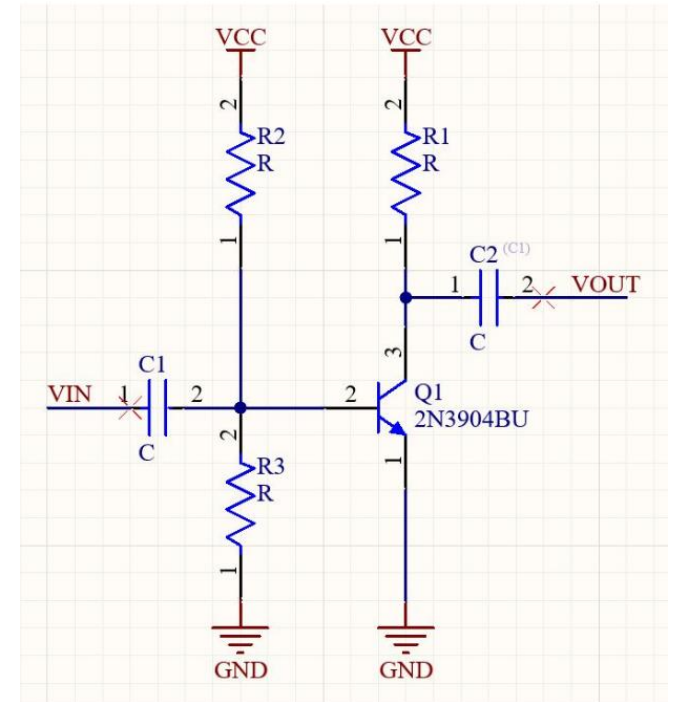
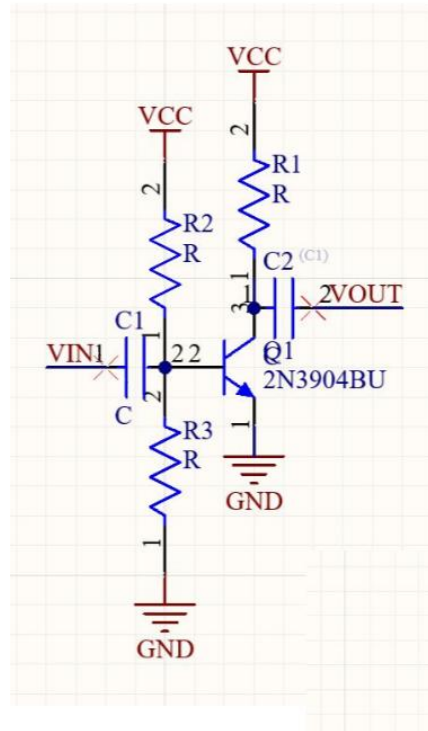
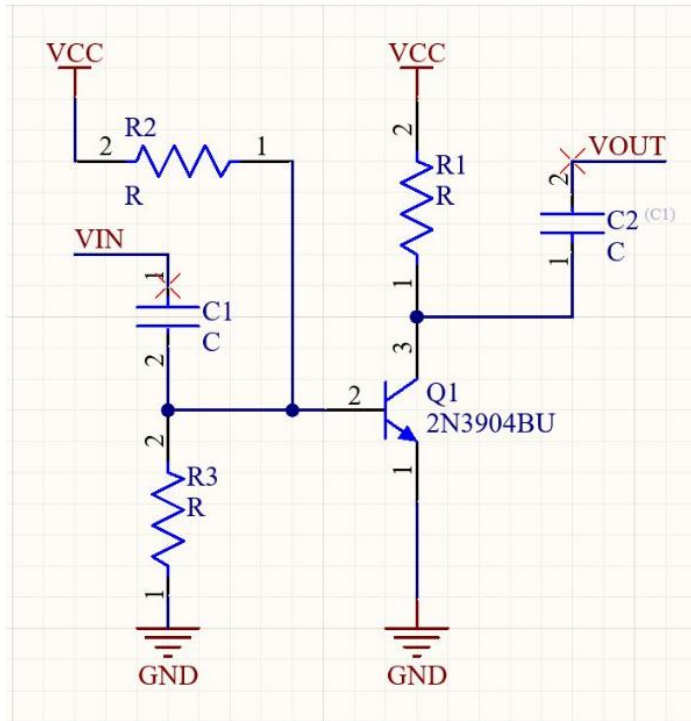


<https://xkcd.com/730/>

Tips:

- Read Left to Right
- Don't Overcrowd
 - Use a larger sheet or use multi-sheet drawings
- Use Logical Blocks to Divide Circuit Elements
- Reference Manufacturer Datasheets
 - Follow established layouts
- Label Signals
 - Especially those which are not immediately obvious
- Keep Power Sources at the top and GND at the Bottom

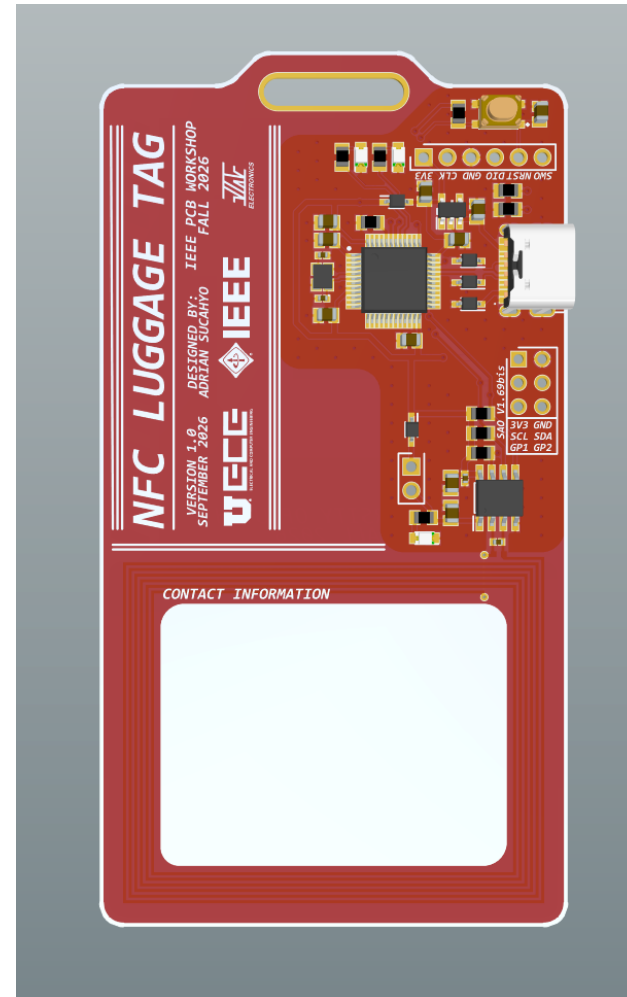
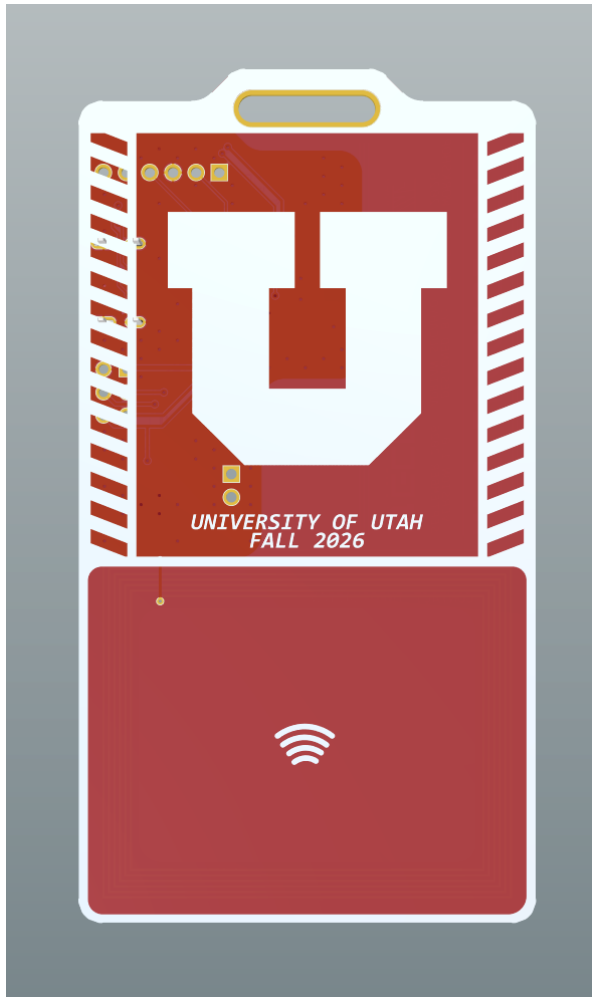
Examples:



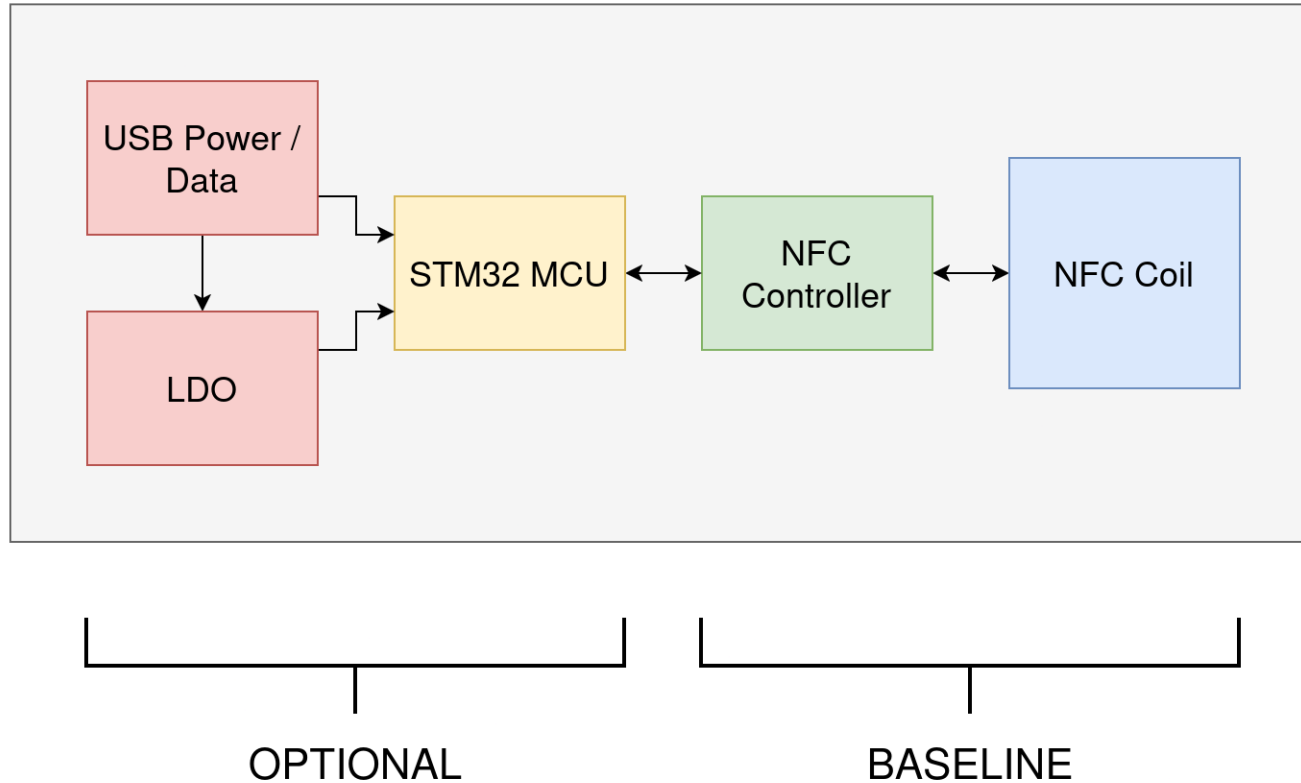
Planning and Execution

- 1. Define what should be achieved
 - The project scope should be defined before beginning
 - Protocols (I2C, UART)
 - Peripherals (Temperature Sensing, Displays)
 - Power Considerations (Low Power, Battery)
 - Block Diagram!
 - Helps to limit what should and should not be designed
- 2. Research how it should be achieved
 - Research what components can achieve your goals
 - Document! Don't forget what you've already seen
- 3. Implement
 - Using researched components to implement project definitions
 - Iterative! If something doesn't work – go back

Project Introduction



Project Block Diagram



NFC Controller and Coil

- **NFC Compatible IC**
 - Integrated Storage
 - Energy Harvesting
 - Select: ST25DV16KC
- **NFC Coil**
 - PCB Coil for Simplicity
 - Estimate coil inductance and tune resonance for 13.56 MHz



ST25DV04KC ST25DV16KC ST25DV64KC

Datasheet

Dynamic NFC/RFID tag IC with 4-, 16-, or 64-Kbit EEPROM, fast transfer mode capability, and optimized I²C



Product status
ST25DV04KC
ST25DV16KC
ST25DV64KC

Features

Includes ST state-of-the-art patented technology

PC interface

- Two-wire I²C serial interface supports 1 MHz protocol
- Single supply voltage: 1.8 to 5.5 V
- Multiple byte programming (up to 256 bytes)
- Configurable I²C target address

Contactless interface

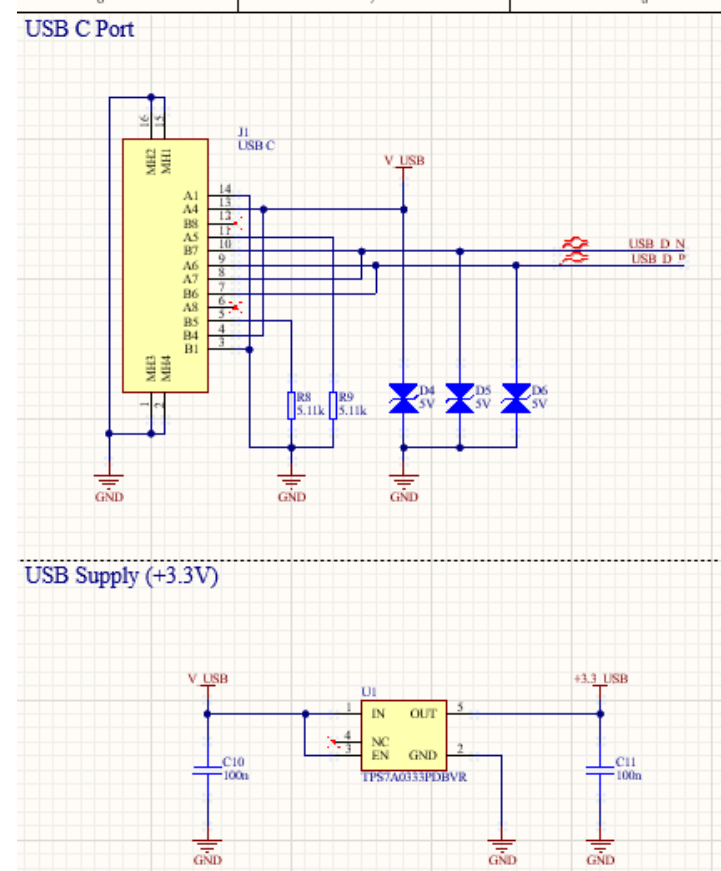
- Based on ISO/IEC 15693
- NFC Forum Type 5 tag certified by the NFC Forum
- Support of all ISO/IEC 15693 modulations, coding, subcarrier modes and data rates
- Custom fast read access up to 53 Kbit/s
- Single and multiple blocks read (same for extended commands)
- Single and multiple (up to four) blocks write (same for extended commands)
- Internal tuning capacitance: 28.5 pF

Memory

- Up to 64 Kbit of EEPROM (depending on version)
- I²C interface accesses bytes
- RF interface accesses blocks of 4 bytes
- Write time:
 - From I²C: typical 5 ms for 1 up to 16 bytes
 - From RF: typical 5 ms for one block
- Data retention: 40 years
- Write cycles endurance:
 - 1 million at 25 °C
 - 600k at 85 °C
 - 500k at 105 °C
 - 400k at 125 °C

USB Power and Regulator (Optional)

- **USB-C Power Input**
 - 5V Input Supply
 - ESD Protection
 - Power MCU and NFC Coil for test and verification
- **3.3V LDO Regulator**
 - 5V USB Input Supply regulated to 3.3V



ECAD Software

- ECAD (Electronic Computer-Aided Design) are software used to design and create diagrams and layouts for PCBs
- There are many different vendors for ECAD software:
 - Altium
 - KiCAD
 - Eagle
 - OrCAD
 - EasyEDA

Altium
Designer®

KiCad

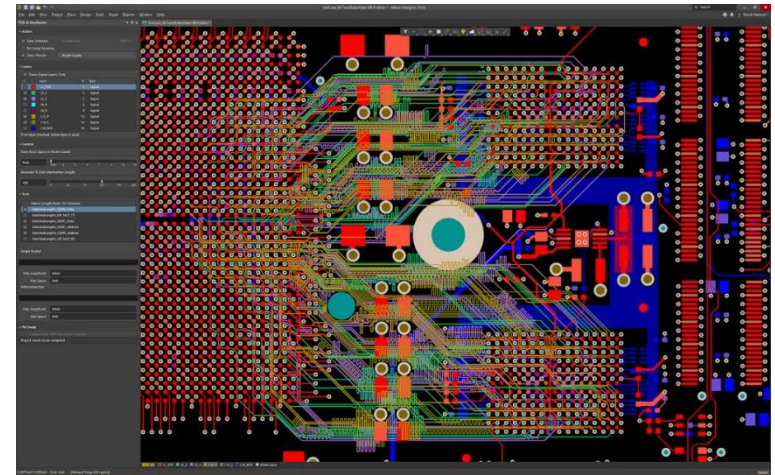
E AUTODESK®
EAGLE

OrCAD™
CADENCE PCB SOLUTIONS

 **EasyEDA**

Why Altium?

- Altium is very commonly used in the industry for PCB and circuit assembly
- Not many opportunities to get access to the software outside of university
 - Please get your student license ASAP if you haven't already
- Great addition to your resume!



EMPLOYMENT / JOB APPLICATION

PERSONAL INFORMATION

FULL NAME: _____ DATE: _____
First Middle Last

ADDRESS: _____
Street Address Apt/Suite

City State Zip Code

E-MAIL: _____ PHONE: _____

SOCIAL SECURITY NUMBER (SSN): ____ - ____ - ____

DATE AVAILABLE: _____ DESIRED PAY: \$ _____ ☐ HOUR ☐ SALARY

POSITION APPLIED FOR: _____

EMPLOYMENT DESIRED: ☐ FULL-TIME ☐ PART-TIME ☐ SEASONAL

Why KiCAD?

- KiCAD is an open source ECAD software alternative!
- Its a great Altium alternative as it doesn't require a license to use
- Some Clubs and Companies use KiCAD as their ECAD software
- Another great addition to your resume!

Questions?

Questions?

Download Today's Project Files

Navigate to the workshop GitHub and
download or clone today's files

<https://github.com/IEEE-U-of-U/IEEE-PCB-Workshop-Fall-2026>